TEST REPORT
ETSI EN 301 489-1 V2.2.3 (2019-11)
ETSI EN 301 489-3 V2.1.1 (2019-03)

Report Reference No.....:	**OCET221020001EC-Y1**
Supervised by (position+printed name+signature)..:	Test Engineer/Mark Tan
Approved by (position+printed name+signature)..:	Manager/Jack Zhang
Date of issue.....:	Oct 26, 2022
Testing Laboratory Name.....:	Shenzhen Eurotest Certification Co.,Ltd
Address.....:	3F, Huafeng Business Building, Pu'an Industrial Zone, Bao'an District,Shenzhen, China
Applicant's name.....:	SHENZHEN JUTAI COMM CO., LTD
Address.....:	5F,MingJinHai Industrial Park,ShiyanTown, BanAn, Shenzhen ,China
Test specification.....:	
Standard.....:	**ETSI EN 301 489-1 V2.2.3 (2019-11)** **ETSI EN 301 489-3 V2.1.1 (2019-03)**

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Test item description.....:	Industry Opening Sensor.
Trade Mark.....:	JUTAI
Manufacturer.....:	SHENZHEN JUTAI COMM CO., LTD
Model/Type reference.....:	RVP42-9M
List Model.....:	RVP42-9ME, RVP42-9MG, RVP42-6S, RVP42-12MS
Hardware version.....:	ASK
Software version.....:	24.165GHZ
Ratings.....:	DC 5V, 100mA
Result.....:	**PASS**

TEST REPORT

Equipment under Test	:	Industry Opening Sensor.
Model /Type	:	RVP42-9M
Listed Models	:	RVP42-9ME, RVP42-9MG, RVP42-6S, RVP42-12MS
Applicant	:	SHENZHEN JUTAI COMM CO., LTD
Address	:	5F,MingJinHai Industrial Park,ShiyanTown, BanAn, Shenzhen ,China
Manufacturer	:	SHENZHEN JUTAI COMM CO., LTD
Address	:	5F,MingJinHai Industrial Park,ShiyanTown, BanAn, Shenzhen ,China

Test Result	**PASS**

The test report merely corresponds to the test sample.
It is not permitted to copy extracts of these test result without the written permission of the test laboratory.

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1. TEST STANDARDS

The tests were performed according to following standards:

ETSI EN 301 489-1 V2.2.3 (2019-07)–ElectroMagnetic Compatibility (EMC) standard for radio equipment and services; Part 1: Common technical requirements; Harmonised Standard for ElectroMagnetic Compatibility
ETSI EN 301 489-3 V2.1.1 (2019-03)–ElectroMagnetic Compatibility (EMC) standard for radio equipment and services; Part 3: Specific conditions for Short-Range Devices (SRD) operating on frequencies between 9 kHz and 246 GHz; Harmonised Standard covering the essential requirements of article 3.1(b) of Directive 2014/53/EU

2. SUMMARY

2.1. General Remarks

Date of receipt of test sample	:	Oct. 18, 2022
Testing commenced on	:	Oct. 18, 2022
Testing concluded on	:	Oct. 26, 2022

2.2. Product Description

Product Name:	Industry Opening Sensor.	
Model/Type reference:	RVP42-9M, RVP42-9ME, RVP42-9MG, RVP42-6S, RVP42-12MS	
Power supply:	DC 5V	
Wireless technology		
Frequency band:	24.165GHZ	
Operating frequency:	24.165GHZ	
Modulation type:	ASK	
Antenna type:	Built-in antenna	
ANT Gain	0.0dBi	
Spread spectrum method:	☒ Duty cycle ☐ Polite spectrum access	
Receiver category:	☐ Category 1:	Category 1 is a high performance level of receiver. In particular to be used where the operation of a SRD may have inherent safety of human life implications.
	☐ Category 1.5:	Category 1.5 is an improved performance level of receiver category 2.
	☒ Category 2:	Category 2 is standard performance level of receiver.
	☐ Category 3:	Category 3 is a low performance level of receiver. Manufacturers have to be aware that category 3 receivers are not able to work properly in case of coexistence with some services such as a mobile radio service in adjacent bands. The manufacturer shall provide another mean to overcome the weakness of the radio link or accept the failure.

2.3. EUT operation mode

Test mode	ASK
1	■

Note:
1. ■ is operation mode.
2. Pre-scan above all test mode, found below test mode which it was worse case mode. Test results reported represents the worst case simultaneous transmission condition.

Test item	Test mode (Worse case mode)
Conducted emission（AC Mains）	Mode 1
Radiated emission	Mode 1
EMS	Mode 1

2.4. EUT configuration

The following peripheral devices and interface cables were connected during the measurement:

○ - supplied by the manufacturer
○ - supplied by the lab

○	DC calbe	Length	<1.2m

2.5. Modifications

No modifications were implemented to meet testing criteria.

3. TEST ENVIRONMENT

3.1. Address of the test laboratory

Shenzhen Eurotest Certification Co.,Ltd
3F, Huafeng Business Building, Pu'an Industrial Zone,
Bao'an District,Shenzhen, China

3.2. Environmental conditions

During the measurement the environmental conditions were within the listed ranges:

Normal Temperature:	25°C
Lative Humidity	55 %
Air Pressure	989 hPa

3.3. Test Description

Emission Measurement		
Radiated Emission	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-3 V2.1.1 (2019-03)	PASS
Conducted Emission(DC Mains)	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-3 V2.1.1 (2019-03)	N/A
Harmonic Current Emissions	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-3 V2.1.1 (2019-03)	N/A
Voltage Fluctuations and Flicker	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-3 V2.1.1 (2019-03)	N/A
Immunity Measurement		
Electrostatic Discharge	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-3 V2.1.1 (2019-03)	PASS
RF Electromagnetic Field	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-3 V2.1.1 (2019-03)	PASS
Fast Transients Common Mode	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-3 V2.1.1 (2019-03)	N/A
RF Common Mode 0,15 MHz to 80 MHz	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-3 V2.1.1 (2019-03)	N/A
Voltage Dips and Interruptions	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-3 V2.1.1 (2019-03)	N/A
Surges	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-3 V2.1.1 (2019-03)	N/A

Remark:
1. N/A means "not applicable".
2.The measurement uncertainty is not included in the test result.

3.4. Statement of the measurement uncertainty

The data and results referenced in this document are true and accurate. The reader is cautioned that there may be errors within the calibration limits of the equipment and facilities. The measurement uncertainty was calculated for all measurements listed in this test report acc. to CISPR 16 - 4 „Specification for radio disturbance and immunity measuring apparatus and methods – Part 4: Uncertainty in EMC Measurements"and is documented in the **Shenzhen Qstlab Testing Co., Ltd.**
acc. to DIN EN ISO/IEC 17025. Furthermore, component and process variability of devices similar to that tested may result in additional deviation. The manufacturer has the sole responsibility of continued compliance of the device.
Hereafter the best measurement capability for **Shenzhen Qstlab Testing Co., Ltd.**

for Products Quality is reported:

Test	Range	Measurement Uncertainty	Notes
Radiated Emission	30~1000MHz	/	(1)
Radiated Emission	1~18GHz	5.16 dB	(1)
Conducted Disturbance	0.15~30MHz	/	(1)

(1) This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=2.

3.5. Equipments Used during the Test

Item	Test Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Calibration Due Date
1	EMI Test Receiver	R&S	ESPI 3	101841-cd	2022/02/24	2023/02/25
2	Artificial Mains	ROHDE & SCHWARZ	ESH2-Z5	893606/008	2022/02/24	2023/02/25
3	Pulse Limiter	Agilent	11947A	3107A04120	2022/02/24	2023/02/25
4	EMI Test Receiver	ROHDE & SCHWARZ	ESCI 7	101102	2022/02/24	2023/02/25
5	Spectrum Analyzer	Agilent	N9020A	MY48010425	2022/02/24	2023/02/25
6	Spectrum Analyzer	R&S	FSV40	100019	2022/02/24	2023/02/25
7	Active Loop Antenna	Beijing Da Ze Technology Co.,Ltd.	ZN30900C	15006	2022/02/24	2023/02/25
8	By-log Antenna	SCHWARZBECK	VULB9163	000976	2022/02/24	2023/02/25
9	Double Ridged Horn Antenna (1~18GHz)	SCHWARZBECK	BBHA 9120D	01622	2022/02/24	2023/02/25
10	Horn Antenna (18GHz~40GHz)	Schwarzbeck	BBHA9170	791	2022/02/24	2023/02/25
11	Amplifier (30MHz~1GHz)	Schwarzbeck	BBV 9743	#202	2022/02/24	2023/02/25
12	Amplifier (1GHz~18GHz)	Taiwan Chengyi	EMC051845B	980355	2022/02/24	2023/02/25
13	Amplifier (26.5GHz~40GHz)	Schwarzbeck	BBV9179	9719-025	2022/02/24	2023/02/25
14	ESD Simulators	EMC Partner	ESD3000	ESD3000-1680	2022/02/24	2023/02/25
15	RF POWER AMPLIFIER	OPHIR	5225R	1079	2022/02/24	2023/02/25
16	RF POWER AMPLIFIER	OPHIR	5273F	1025	2022/02/24	2023/02/25
17	RF POWER AMPLIFIER	SKET	HAP_0306G-50W	--	2022/02/24	2023/02/25
18	Stacked Broadband Log Periodic Antenna	SCHWARZBECK	STLP 9128	9128ES-112	2022/02/24	2023/02/25
19	Stacked Mikrowellen Log.-Per Antenna	SCHWARZBECK	STLP 9149	9149-594	2022/02/24	2023/02/25
20	Electric field probe	Narda S.TS./PMM	EP601	611WX90121	2022/02/24	2023/02/25
21	ESG Vector Signal Generator	Agilent	MY49072627	E4594C	2022/02/24	2023/02/25
22	Power meter	Agilent	MY45104493	E4602B	2022/02/24	2023/02/25
23	Power sensor	Agilent	MY41495234	E7201H	2022/02/24	2023/02/25
24	Ultra Compact Simulator	EMC Partner	TRANSIENT3000	TRA3000 F5-S-D-V-1527	2022/02/24	2023/02/25
25	Coupling Clamp	EMC Partner	CN-EFT1000	CN-EFT1000-1574	2022/02/24	2023/02/25
26	Signal Line Coupling Network	EMC Partner	CN-R40C05	CN-R40C05-1513	2022/02/24	2023/02/25

27	CS Test system	Frankonia	CIT-10-75	126B1333	2022/02/24	2023/02/25
28	6dB Attenuator	Frankonia	75-A-FFN-06	1509	2022/02/24	2023/02/25
29	CDN	Frankonia	M2+M3	A2210239	2022/02/24	2023/02/25
30	Power Clamp	Frankonia	EMCL-20	132A1216	2022/02/24	2023/02/25
31	Harmonic and Flicker Analyzer	EMC Partner	HARMONICS 1000	HAR1000-1P 230V-0221	2022/02/24	2023/02/25
32	RF Cable	HUBER+SUHNER	RG214	N/A	2022/02/24	2023/02/25
33	Conducted Emission	Tonscend	JS32-CE	Ver 2.5	/	/
34	Radiated Emission	Tonscend	JS32-RE	Ver 2.5.1.8	/	/

The calibration interval is 1 year.

4. TEST CONDITIONS AND RESULTS

4.1. EN/ASSION

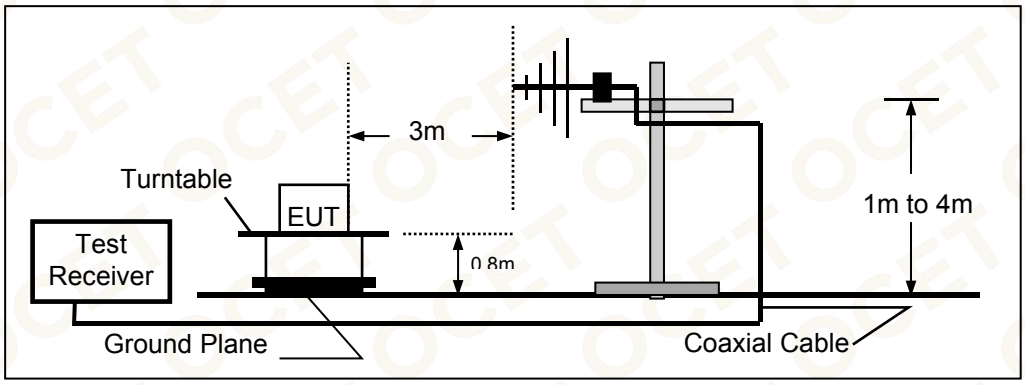
4.1.1. Radiated Emission

LIN/AT

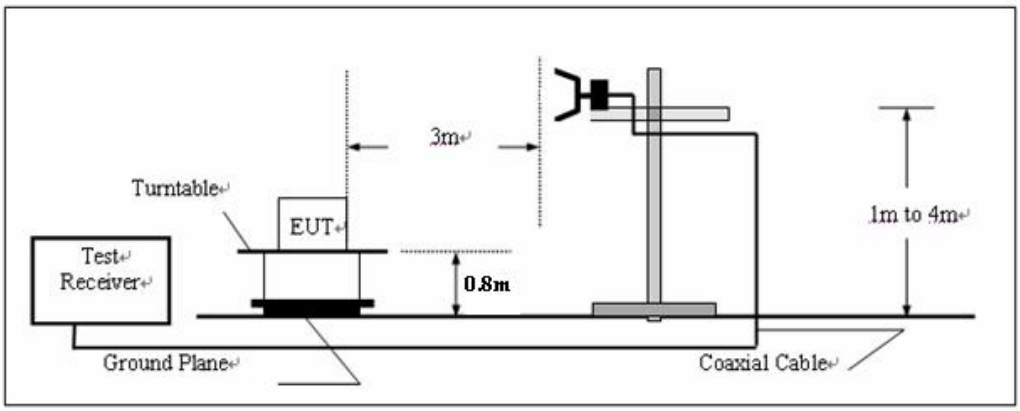
Please refer to ETSI EN301489-1 Clause 8.2.3, Table 4 and EN55032 Annex A, Table A.2,A.3, and Class B

TEST CONFIGURATION

a) Radiated emission test set-up, frequency below 1000MHz:



b) Radiated emission test set-up, frequency above 1000MHz



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 8.2.3 and EN55032 Annex A for the measurement methods

TEST RESULTS

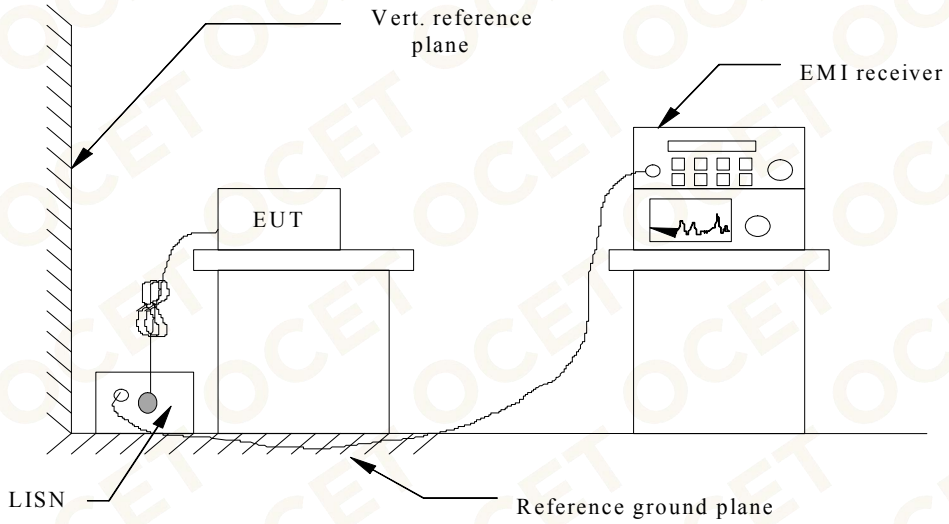
Passed

4.1.2. Conducted Emission

LIN/AT

Please refer to ETSI EN301489-1 Clause 8.4.3, Table 8 and EN55032 Annex A, Table A.10, A.12

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 8.4.3 and EN55032 Annex A for the measurement methods.

TEST RESULTS

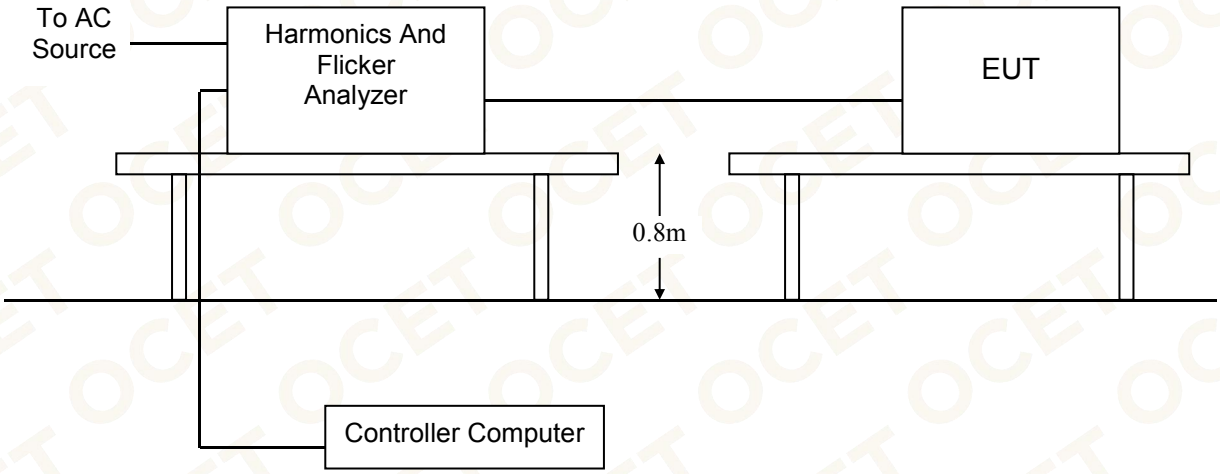
N/A

4.1.3. Harmonic Current Emission

LIN/AT

Please refer to EN 61000-3-2

TEST CONFIGURATION



TEST PROCEDURE

Please refer to EN 61000-3-2 for the measurement methods.

TEST RESULTS

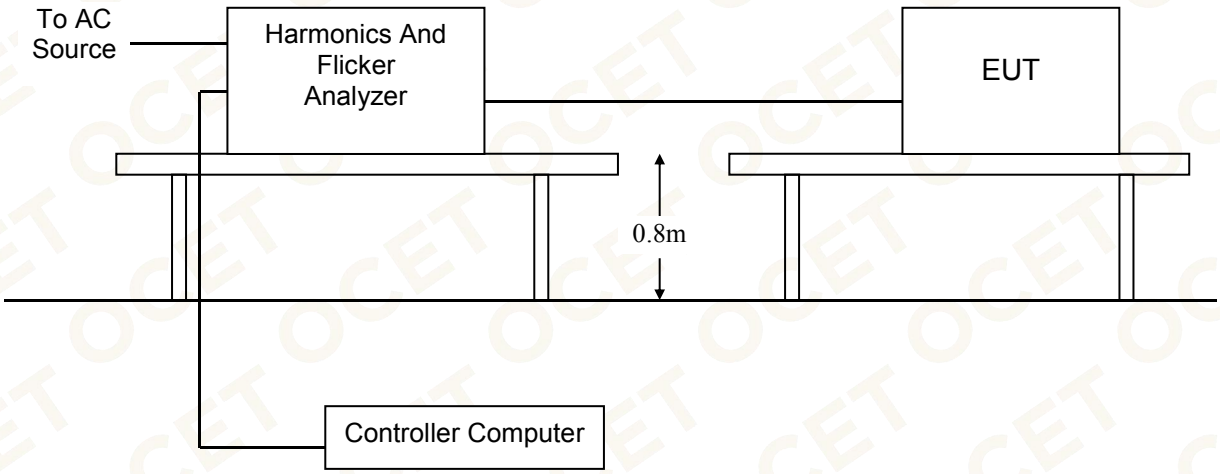
N/A

4.1.4. Voltage Fluctuation and Flicker

LIN/AT

Please refer to EN 61000-3-3

TEST CONFIGURATION



TEST PROCEDURE

Please refer to EN 61000-3-3 for the measurement methods.

TEST RESULTS

N/A

4.2. IMMUNITY

4.2.1. Performance criteria

■ **ETSI EN301489-3**

General performance criteria

• performance criterion A applies for immunity tests with phenomena of a continuous nature;

• performance criterion B applies for immunity tests with phenomena of a transient nature.

•The equipment shall meet the minimum performance criteria as specified in the following.

Criteria	During test	After test
A	Operate as intended No loss of function No unintentional responses	Operate as intended No loss of function No degradation of performance No loss of stored data or user programmable functions
B	May show loss of function No unintentional responses	Operate as intended Lost function(s) shall be self-recoverable No degradation of performance No loss of stored data or user programmable functions
NOTE: Where "operate as intended" or "no loss of function" is specified, the EUT shall demonstrate correct functioning as described in clause 5.		

4.2.2. Monitoring EUT in Immunity Test

4.2.2.1 Monitoring for Continuous Phenomena Applied to the EUT

■ **433M Transmitter and Receiver**

Before the test starts, Press Power Button of the transmitter, the Receiver work as intended;

During the test, observe whether the EUT working as intended and no unintentional responses by eyes or monitor;

After the test, Check the function and no degradation of performance.

4.2.2.2 Monitoring for Transient Phenomena Applied to the EUT

■ **433M Transmitter and Receiver**

Before the test starts, Press Power Button of the transmitter, the Receiver work as intended;

After the test, Check the function and no degradation of performance

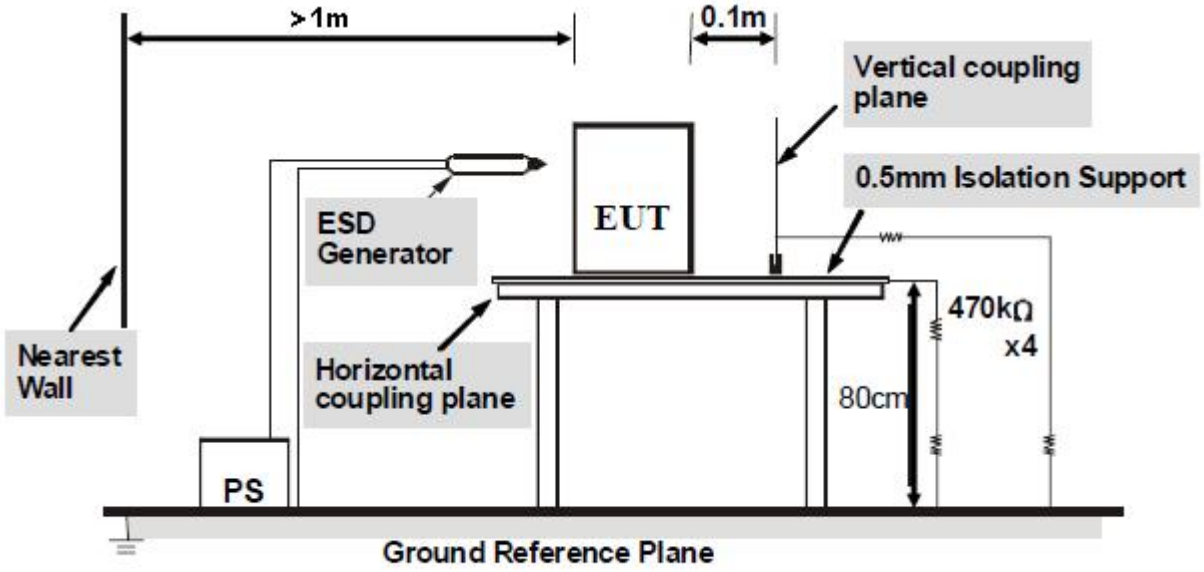
4.2.3. Electrostatic Discharge

LIN/AT

SEVERITY LEVELS OF ELECTROSTATIC DISCHARGE

Test level: Contact Discharge at ±2KV, ±4KV Air Discharge at ±2KV, ±4KV, ±8KV

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 9.3.2 , EN 55035 and EN 61000-4-2 for the measurement methods.

Contact Discharge:

The ESD generator is held perpendicular to the surface to which the discharge is applied and the tip of the discharge electrode touch the surface of EUT. Then turn the discharge switch. The generator is then re-triggered for a new single discharge and repeated at least 10 times for each pre-selected test point. This procedure shall be repeated until all the air discharge completed.

Air Discharge:

Air discharge is used where contact discharge can't be applied. The round discharge tip of the discharge electrode shall be approached as fast as possible to touch the EUT. After each discharge, the discharge electrode shall be removed from the EUT. The generator is then re-triggered for a new single discharge and repeated at least 10 times for each pre-selected test point. This procedure shall be repeated until all the air discharge completed.

Indirect discharge for horizontal coupling plane:

At least 10 single discharges shall be applied to the horizontal coupling plane, at points on each side of the EUT.

Indirect discharge for vertical coupling plane:

At least 10 single discharges shall be applied to the center of one vertical edge of the coupling plane. The coupling plane, of dimensions 0.5m X 0.5m, is placed parallel to, and positioned at a distance of 0.1m from the EUT. Discharges shall be applied to the coupling plane, with this plane in sufficient different positions that the four faces of the EUT are completely illuminated.

TEST MODE

Please reference to the section 2.3

TEST RESULTS

Direct discharge				
Type of discharge	Discharge voltage (KV)	Observations Performance	Criteria Level	Result
Contact discharge	±2	No degradation in performance of the EUT was observed (A)	B	Pass
	±4	A	B	
Air discharge	±2	A	B	
	±4	A	B	
	±8	A	B	
Indirect discharge				
Type of discharge	Discharge voltage (KV)	Observations Performance	Criteria Level	Result
HCP (6 sides)	±2	A	B	Pass
	±4	A	B	
VCP (4 sides)	±2	A	B	
	±4	A	B	

Remark: The ancillary equipment's specification for an acceptable level of performance or degradation of performance during and/or after the ESD tests.

4.2.4. RF Electromagnetic Field

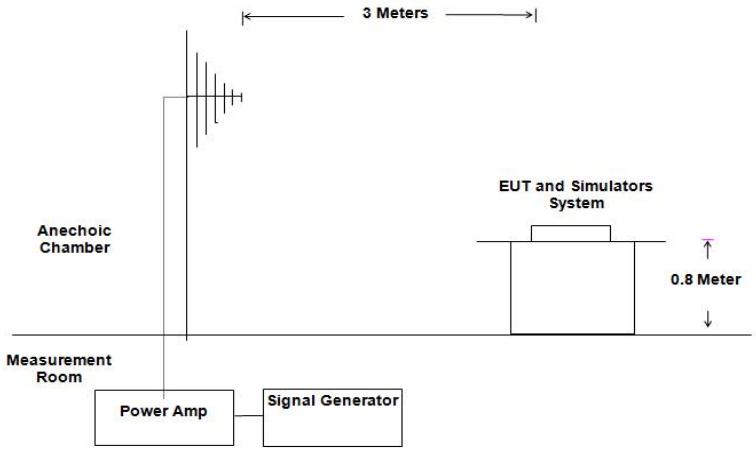
PERFORMANCE CRITERION

Criteria A

TEST LEVEL

3V/m (80%, 1kHz Amplitude Modulation)

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 9.2.2 and EN 61000-4-3 for the measurement methods.

TEST MODE

Please reference to the section 2.3

TEST RESULTS

Frequency	Level	Modulation	Antenna Polarization	EUT Face	Observations (Performance Criterion)	Result
80MHz-25GHz	3 V/m	1 kHz, 80 % Amp. Mod, 1 % increment, dwell time=3seconds	V	Front	A	Pass
			H		A	Pass
			V	Rear	A	Pass
			H		A	Pass
			V	Left	A	Pass
			H		A	Pass
			V	Right	A	Pass
			H		A	Pass
			V	Top	A	Pass
			H		A	Pass
			V	Bottom	A	Pass
			H		A	Pass

Remark: A: No degradation in performance of the EUT was observed.

4.2.5. Surges

PERFORMANCE CRITERION

Criteria B

TEST LEVEL

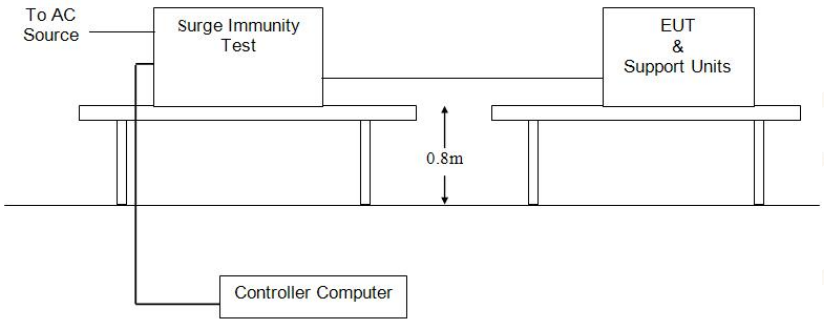
1kV Line to Line: Differential mode

2kV Line to Ground: Common mode

0.5KV Line to Line: network Wire

(Voltage Waveform: 1.2/50 us; Current Waveform: 8/20 us)

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 9.8.2 and EN 61000-4-5 for the measurement methods.

TEST MODE

Please reference to the section 2.3

TEST RESULTS

N/A

4.2.6. RF- Common Mode 0.15MHz to 80MHz

PERFORMANCE CRITERION

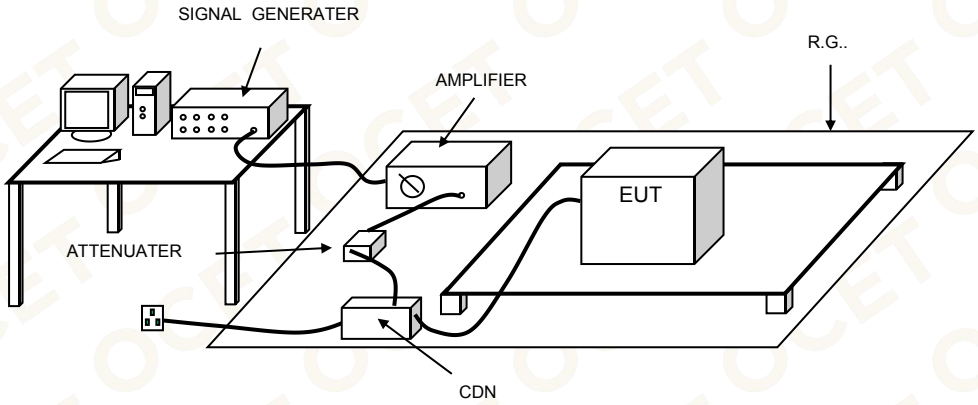
Criteria A

TEST LEVEL

3Vrms on AC main port (80%, 1kHz Amplitude Modulation)

3Vrms on network port (80%, 1kHz Amplitude Modulation)

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 9.5.2 and EN 61000-4-6 for the measurement methods.

TEST MODE

Please reference to the section 2.3

TEST RESULTS

N/A

4.2.7. Fast Transients Common Mode

PERFORMANCE CRITERION

Criteria B

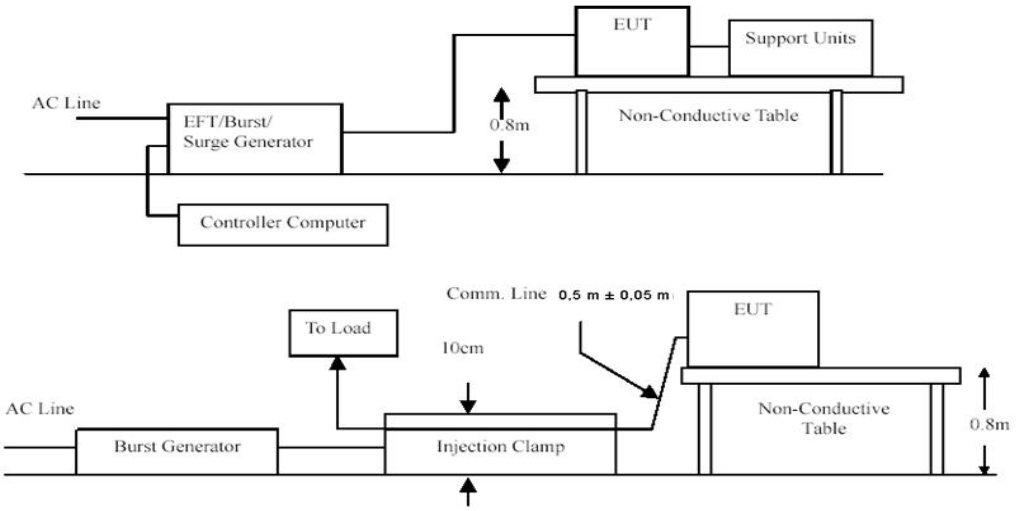
TEST LEVEL

1KV for AC main port

0.5KV for network port

(Impulse Frequency: 5 kHz; Tr/Th: 5/50ns; Burst Duration: 15ms; Burst Period: 3Hz)

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 9.4.2， EN55035 and EN 61000-4-4 for the measurement methods.

TEST MODE

Please reference to the section 2.3

TEST RESULTS

N/A

4.2.8. Voltage Dips and Interruptions

PERFORMANCE CRITERION

>95% VD, 0.5 period----Performance criterion: B

>95% VD, 1.0 period----Performance criterion: B

30% VD, 25 period----Performance criterion: C

>95% VI, 250 period----Performance criterion: C

TEST LEVEL

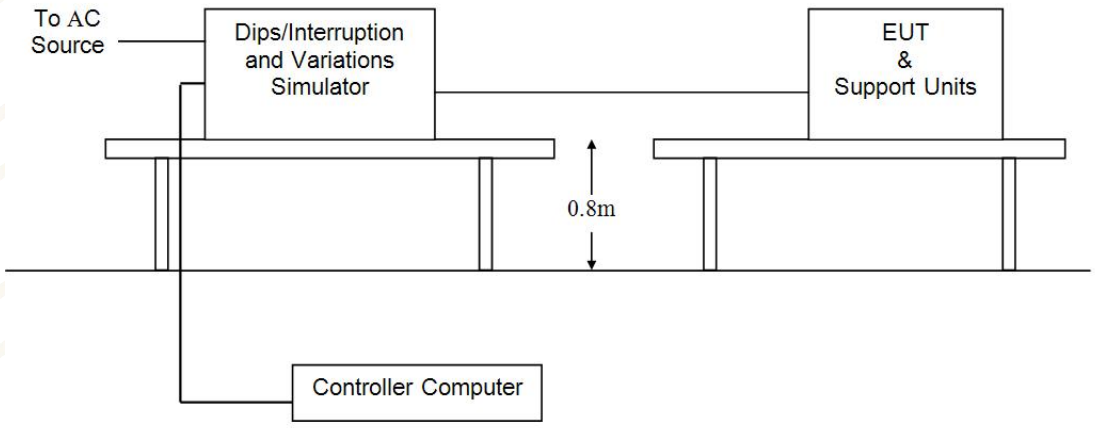
0% of VT(Supply Voltage) for 0.5 period

0% of VT(Supply Voltage) for 1.0 period

70% of VT(Supply Voltage) for 25 period

0% of VT(Supply Voltage) for 250 period

TEST CONFIGURATION



TEST PROCEDURE

Please refer to ETSI EN 301 489-1 Clause 9.7.2 and EN 61000-4-11 for the measurement methods.

TEST MODE

Please reference to the section 2.3

TEST RESULTS

N/A

5. External and Internal Photos of the EUT

Photo 1



Photo 2



Photo 3



Photo 4



Photo 5



.....End of Report.....